# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Machine Learning

## ASSESSMENT TOOL 1 – Data Interpretation

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|---|---|---|---|
| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO4 – Precision | **Assessment:** | Assessment Tool 1 – Data Interpretation |
| **Weightage:** | 40% of CIA | | |
| **CO5 Statement:** | Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4) | | |
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| **Section / Batch:** | AI & ML / 2024 | **Date:** | 10/08/2026 |

**Code of Conduct**

*I __D. Sri Anjaneya / 192425219__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: __D. Sri Anjaneya__

**Instructions**

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

| Section / Topic | Focus | Question No. |
|---|---|---|
| Representing Meaning & Meaning Structure of Language | Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems. | Q1 |
| First-Order Predicate Calculus & Representing Linguistically Relevant Concepts | Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments. | Q2 |
| Lexemes and Their Senses & Word Sense Disambiguation | Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems. | Q3 |

| | | |
|---|---|---|
| Syntax-Driven Semantic Analysis | Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications. | Q4 |

## Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
   A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

| Customer Query | Semantic Representation |
|---|---|
| "Activate international roaming for my number." | ACTIVATE(Roaming, Customer) |
| "Deactivate caller tune service." | DEACTIVATE(CallerTune, Customer) |
| "Check my data balance." | QUERY(DataBalance, Customer) |
| "Enable 5G service." | ACTIVATE(5GService, Customer) |

   Additional chatbot logs show:

| Query ID | Actual User Intent | Predicted Intent |
|---|---|---|
| Q1 | Activate Roaming | Activate Roaming |
| Q2 | Deactivate Caller Tune | Activate Caller Tune |
| Q3 | Check Data Balance | Query Data Balance |
| Q4 | Enable 5G Service | Activate 5G Service |

   **Tasks:**

   1. Analyze the semantic representations and identify the action-object relationships.
   2. Determine which query contains semantic interpretation errors.
   3. Evaluate how meaning representation affects chatbot decision accuracy.
   4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
   An Industry 4.0 manufacturing plant monitors machines using semantic rules. Production Data:

| Machine | Status |
|---|---|
| M1 | Active |
| M2 | Active |
| M3 | Maintenance |
| M4 | Active |

   **Predicate Rules:**

- Active(x) → Producing(x)
- Produces(x,y)∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x)

   **Tasks:**

   1. Represent the production data using First-Order Predicate Calculus.

2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines. An e-commerce company observed the following search logs.

| User Query | Possible Sense 1 | Possible Sense 2 |
|---|---|---|
| "Apple accessories" | Fruit | Technology Brand |
| "Mouse wireless" | Animal | Computer Device |
| "Java tutorial" | Island | Programming Language |
| "Python course" | Snake | Programming Language |

Search Context Data:

| Query | Clicked Result |
|---|---|
| Apple accessories | iPhone Charger |
| Mouse wireless | Bluetooth Mouse |
| Java tutorial | Coding Lessons |
| Python course | Software Development Training |

**Tasks:**

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

| Sentence | Parse Structure |
|---|---|
| Doctor prescribed medicine to patient. | Subject-Verb-Object |
| Patient reported severe headache. | Subject-Verb-Object |
| Nurse monitored patient continuously. | Subject-Verb-Object |
| Medicine reduced blood pressure. | Subject-Verb-Object |

Semantic Roles Generated:

| Entity | Role |
|---|---|
| Doctor | Agent |
| Medicine | Instrument |
| Patient | Recipient |
| Headache | Symptom |

**Tasks:**

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

# 1. Semantic Representation in Customer Support Chatbots

### 1. Action–Object Relationships

The semantic representations follow the structure **Action (Object, Entity)**.

- **ACTIVATE (Roaming, Customer)** → Action: Activate; Object: Roaming.
- **DEACTIVATE (CallerTune, Customer)** → Action: Deactivate; Object: Caller Tune.
- **QUERY (DataBalance, Customer)** → Action: Query; Object: Data Balance.
- **ACTIVATE (5GService, Customer)** → Action: Activate; Object: 5G Service.

Thus, the representation identifies **what action the customer wants and which service is involved**.

### 2. Semantic Interpretation Error

- Actual intent: **Deactivate Caller Tune**
- Predicted intent: **Activate Caller Tune**

The system correctly identified **Caller Tune** but incorrectly interpreted **Deactivate** as **Activate**. Since these actions have opposite meanings, the chatbot could perform the wrong operation.

### 3. Effect on Chatbot Decision Accuracy

Meaning representation directly affects chatbot decision-making. A correct representation such as:

**ACTIVATE(Roaming, Customer)**

allows the chatbot to execute the correct service operation. An incorrect action can cause the system to perform the opposite operation.

Therefore, accurate semantic representation improves **intent classification, service execution and customer satisfaction**.

### 4. Recommended Improvements

1. Use larger telecom-specific training datasets.
2. Train the system to distinguish opposite actions such as activate/deactivate and enable/disable.

3. Use previous conversation context to understand customer intent.

4. Add confidence scores and confirmation prompts for sensitive service operations.

**Conclusion:** Accurate semantic representation is essential because even a small error in the action can result in an incorrect telecom service operation.

# 2. First-Order Predicate Calculus for Smart Manufacturing

### 1. Production Data Using First-Order Predicate Calculus

The production data can be represented as:

The given rules are:

Therefore:

- M1 → Active → Producing
- M2 → Active → Producing
- M3 → Maintenance → Not Producing
- M4 → Active → Producing

### 2. Products Currently Available

The rule:

requires information about which machine produces which product.

The given data does **not specify Produces(Machine, Product) relationships**. Therefore, the exact products currently available cannot be determined.

For example, if:

and:

then:

### 3. Is Gear Production Affected by Maintenance?

M3 is under maintenance:

However, the data does not state whether **M3 produces Gear**.

Therefore, Gear production cannot be confirmed as affected.

If:

then Gear production would be affected because M3 is not producing.

**4. Effectiveness of Predicate Logic**

Predicate logic is useful in industrial decision-support systems because it represents machine states and production rules precisely.

- Represents machine conditions formally.
- Supports logical inference.
- Helps identify machines unavailable for production.
- Supports automated production decisions.

**Conclusion:** Predicate logic effectively connects **machine status → production status → product availability**, but complete production facts are required for accurate conclusions.

## 3. Word Sense Disambiguation in E-Commerce Search Engines

### 1. Correct Word Sense for Each Query

The clicked search results provide strong contextual evidence.

| Query | Correct Sense | Evidence |
|---|---|---|
| Apple accessories | Technology Brand | iPhone Charger |
| Mouse wireless | Computer Device | Bluetooth Mouse |
| Java tutorial | Programming Language | Coding Lessons |
| Python course | Programming Language | Software Development Training |

Thus, the search engine should use the surrounding context rather than selecting a meaning based only on the dictionary definition.

### 2. Semantic Cues Used

The important cues are:

- **Accessories** → indicates a technology product in "Apple accessories".
- **Wireless** → indicates a computer mouse.
- **Tutorial** → strongly indicates Java programming.
- **Course/Training** → indicates Python programming.

The clicked results also provide **user-behavioral context** for determining the intended sense.

### 3. Impact of Incorrect Sense Selection

Incorrect word-sense selection can reduce search performance.

For example, interpreting **Python course** as a snake could produce wildlife-related results instead of programming courses.

This can cause:

- Lower search relevance.
- Reduced click-through rate.
- Poor recommendations.
- User dissatisfaction.

**4. Industrial-Scale WSD Strategy**

An e-commerce company can improve WSD by:

1. Using contextual language models to understand query meaning.
2. Combining query context with product titles and categories.
3. Using previous searches and clicks as behavioral information.
4. Continuously updating the model using user feedback and new search data.

**Conclusion:** Combining **linguistic context, product information and user behavior** provides more accurate word-sense disambiguation.

# 4. Syntax-Driven Semantic Analysis in Healthcare

### 1. Role of Syntax in Semantic Role Assignment

Syntactic structure helps identify relationships between entities and actions.

For example:

**"Doctor prescribed medicine to patient."**

- Doctor → Subject
- Prescribed → Verb
- Medicine → Object
- Patient → Recipient

This structure helps the semantic analyzer determine **who performed the action, what was involved and who received it**.

### 2. Appropriateness of Semantic Roles

Some assigned roles are correct, while one needs correction.

| Entity | Given Role | Evaluation |
|---|---|---|
| Doctor | Agent | Correct |
| Medicine | Instrument | Generally incorrect; better as Theme/Object |
| Patient | Recipient | Correct |
| Headache | Symptom | Correct |

For **"Doctor prescribed medicine to patient"**, medicine is better considered the **Theme/Object** rather than an Instrument.

### 3. Errors from Incorrect Parsing

Incorrect parsing can lead to:

- Wrong identification of the Agent.
- Incorrect association between a treatment and patient.
- Incorrect interpretation of medical relationships.
- Errors in clinical information extraction and decision support.

For example, confusing the doctor and patient roles could incorrectly indicate who prescribed the medicine.

**4. Methods to Improve Analysis**

1. Use medical-domain NLP models trained on healthcare documents.
2. Combine dependency parsing with semantic-role labeling.
3. Use healthcare-specific dictionaries and medical ontologies.
4. Use contextual language models to understand entities based on the complete sentence.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Data | 25% | Accurately interprets all data patterns, trends, and relationships | Interprets data correctly with minor gaps | Partial understanding; misses key trends | Misinterprets or fails to understand data |
| Analysis | 25% | Provides deep analysis with strong logical reasoning and justification | Provides reasonable analysis with some justification | Basic analysis with weak reasoning | No meaningful analysis provided |
| Application of Concepts | 20% | Effectively applies relevant concepts/tools to interpret data accurately | Applies concepts with minor errors | Limited application; requires guidance | Unable to apply concepts to data |
| Conclusion | 20% | Draws clear, meaningful conclusions with strong insights and implications | Provides valid conclusions with moderate insights | Conclusions are vague or weak | No clear or valid conclusions |
| Clarity | 10% | Presents interpretation clearly using proper structure, visuals, and terminology | Generally clear with minor issues | Lacks clarity or proper structure | Poor presentation and difficult to understand |

| Q. No. / Criteria Level | Understanding of Data | Analysis | Application of Concepts | Conclusion | Clarity | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
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| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |